

Table of Contents

1	<i>Introduction.....</i>	<i>1</i>
2	<i>Objective</i>	<i>1</i>
3	<i>Participants and Examiner</i>	<i>1</i>
4	<i>List of Tools</i>	<i>1</i>
5	<i>Data Collection Scope.....</i>	<i>2</i>
5.1	<i>Disclaimer regarding data collection</i>	<i>2</i>
6	<i>Methodology</i>	<i>2</i>
6.1	<i>Phase I.....</i>	<i>2</i>
6.2	<i>Phase II</i>	<i>3</i>
7	<i>Expected Outcomes</i>	<i>4</i>
	<i>Appendix -A</i>	<i>i</i>
	<i>Appendix -B</i>	<i>ii</i>

1 Introduction

The growing accessibility of generative AI tools has prompted renewed interest in their potential role within education, particularly in the field of Technology-Integrated Communication in Education (TICE). For modern language teachers, lesson planning represents one of the most cognitively demanding aspects of professional practice, requiring the simultaneous consideration of curriculum objectives, student needs, and pedagogical strategy. AI-assisted planning offers a potential means of reducing this burden while broadening the quality of available teaching resources. This experiment investigates that potential by observing language teachers engage in lesson planning both with and without AI assistance, with the aim of understanding AI not as a replacement for teacher expertise, but as a practical support tool for the new generation of classroom practice.

2 Objective

The objective of this experiment is to understand the advantages of AI in lesson planning for modern language courses.

3 Participants and Examiner

The Experiment Conductor (EC) conducts the experiment in two phases, through instructions, record management, output collection, and providing the necessary tools to the participants.

The participants, individually noted as Test Participant (TP), are a set of eight modern language teachers from a middle school where this experiment is conducted. The selected subjects are French and English. The group of selected teachers teach all four levels at the middle school. These teachers have already participated in the Introductory Questionnaire.

As the participants are middle school teachers who teach either French or English, a topic common to both subjects is selected. The topic is BE + ING / Present Progressive. The participants are required to create a lesson plan for a 2-hour lecture. The topic is the same for both phases; however, it is not revealed to the participants at the beginning of the experiment. The topic is written on a piece of paper and folded, which is provided to the participants when the recording starts.

4 List of Tools

A set of tools are listed that are provided to the TP:

- A stack of A4 plain white sheets.
- A pen.

The following set of tools are used by the EC for recording:

- A camera, if consent is given,
- A timer, if consent for camera is denied,
- A pen,

Both Phase I and Phase II tools are common, except for a laptop with a running internet browser with several tabs open for the following generative AI (Gen-AI) tools:

- ⇒ Chat GPT
- ⇒ Claude AI
- ⇒ Duck AI
- ⇒ Google Gemini

The TP may select any or all of the provided AI tools and select the best lesson plan aide as per output of the AI tool (may combine outputs as well).

And a screen recording application to record all interaction of the TP with the laptop.

The EC provides the laptop with open and running applications and browser tabs during Phase II to the TP.

5 Data Collection Scope

The list of data collection instruments, common for the full experiment, is as follows:

- Consent Form, read and signed by TP.
- 2 x Final Lesson Plan submitted by TP, that must be photographed.
- 2 x Video Recording / Time Recording Table¹ (as per consent)

During Phase II of the experiment, these additional data are recorded, screen recording is additionally captured. The interaction with AI tools are saved as PDF as soon as Phase II is completed.

After the experiment, a Post Test Feedback² questionnaire is filled by the TP. This includes commentaries from the TP regarding the differences noted in the two lesson plans as well as experience of the extent of support provided by AI tools.

5.1 *Disclaimer regarding data collection*

Personal information such as name, age, gender, educational background and experience, video recordings with facial expressions capture are collected during this experiment. These data are required for two aspects:

1. Demographical groupism for test group candidates and their relationship to AI tools usage and experience.
2. Facial expression used to study interactions with AI, to note emotional responses to AI outputs.

The identity and identification marking of the TP are strictly restricted to the EC and will not be revealed at any circumstances. The TPs will be named by random identifiers if specific commentary is made on their experiment. The video recordings will be used for inferences that is required for this report and will be deleted after the final review of this thesis. The video recordings or images will not be shared with anyone under any circumstances.

6 Methodology

The experiment is conducted in two phases, of approximately 20–30 minutes, known as Phase I and Phase II. A few steps are common for both Phase I and Phase II. However, this section is written in a textual flow format of the entire experiment for one TP.

6.1 *Phase I*

The first phase of the experiment is as follows:

1. The EC explains the confidentiality terms to the TP and provides the consent form Appendix - A to be read and signed by TP.
2. The EC asks verbally for consent to video recording, and if not provided, then verbally states the use of timer.
3. The EC explains the procedure to be followed to the TP:
 - a. A topic hidden in the folded piece of paper is provided.
 - b. When EC states verbally “Start” and hands over the piece of paper, the TP receives the topic for the experiment.

¹ To record time steps as well as note emotional response/activity during interaction with Gen-AI.

² Printed hand-out form to be filled by hand.

- c. The TP needs to create a 2-hour lesson plan on the given topic in their respective subject, without any external aide.
 - d. The lesson plan needs to be written on an A4 white plain paper, a stack of papers is made available for this purpose.
 - e. The video recording or timer ends when the TP states verbally “Finished”.
 - f. The lesson plan is to be submitted to the EC.
4. Now, the EC prepares the experiment area to make sure all the tools are available at an arm’s reach.
5. The video recording device or the table to record timer steps and comments must be set up near the EC.
6. The folded piece of paper is with the EC.
7. The EC asks verbally “Ready?” to which the TP is required to reply “Ready”.
8. The EC starts the timer, states “Start” and hands over the folded piece of paper to the TP.
9. The TP unfolds the paper and starts to make a lesson plan for the given topic as required.
10. The EC must note all expressions of the TP and little moments from the timer steps that are relevant to understand the emotion of the TP during this task, if video recording is not permitted.
11. The TP states “Finished” when the lesson plan is ready, and the EC immediately stops the timer and records the final time information.
12. The EC collects the lesson plan, which is required to be photographed.
13. The EC performs a systematic check to ensure all the data is collected without errors. This marks the End of Phase I, for one TP.

6.2 Phase II

The second phase continues after Phase I.

1. The EC provides the instructions for Phase II to the TP:
 - a. The same instructions from 3.a to 3.f are repeated with additional instructions, such as,
 - b. A laptop is provided to the TP with open tabs of selected Gen-AI tools.
 - c. The TP must use one or more of the AI tools to generate a lesson plan for the 2-hour lecture for the given topic. A lesson plan that is created without the help of AI is not acceptable.
 - d. The TP must submit the final lesson plan in A4 sheets of paper.
 - e. The screen is always recorded for the entirety of interaction with the AI tools.
2. The EC continues the experiment as in Phase I. The TP interacts with AI tools, creates a lesson plan and submits the A4 sheet to the EC as in Phase I.
3. The timer/video recording, as well as screen recording collection, PDF of interaction with AI tools and photographs of the lesson plan are verified by the EC, thus marking the end of Phase II.

At the end of Phase II, a feedback form is provided to the TP. Additionally, comments on the differences between the two lesson plans are collected from the TP. These provide the scope for additional inferences on the experience of using AI for lesson planning directly from the participant.

7 Expected Outcomes

- Based on experience the quality of lesson planning differs.
- Difference in expertise with AI may result in various levels of outcomes.
- Ai may help in reducing the preparation time.
- Ai may also provide better more detailed outcomes.
- What AI produced may not correspond to the test participants' goals.

Appendix -A

Participant Information and Consent Form

General Information

Name	
Age	
Last Degree Obtained	
Institution Graduated From (and City, Country)	
Language Currently Teaching	
Years of Experience Teaching Above Language	
Other Languages Taught (Years of Experience)	

Data Protection and Confidentiality

Your identity shall remain strictly confidential throughout the study and in all resulting publications. Personal data is collected solely for research purposes and will not be shared under any circumstances.

Data marked with (*) will be used for statistical analysis and participant profiling only.

All collected data will be securely stored and permanently deleted after thesis presentation.

Video Recording Consent

The experimental session may be video recorded for research analysis purposes. Recordings will be: Accessible only to the experiment conductor and will not be published or shared publicly.

Stored securely and deleted after thesis presentation.

Used exclusively for scientific analysis of the experimental task

☐ Yes I consent to video recording of my experimental session.

☐ No I do not consent to video recording. The experiment will proceed without recording.

Declaration of Informed Consent

I confirm that I have read and understood the information provided above regarding this experiment. The conditions of the study have been explained to me by the experiment conductor, and I have had the opportunity to ask questions. I agree to participate voluntarily under the terms described.

Participant Signature

Date and Place

Appendix -B

Post-Experiment Feedback Form

Generative AI Integration in Teaching Practice

Confidentiality & Ethics Notice

Your responses are strictly confidential and will be used exclusively for this thesis. Data will be stored securely and anonymised before analysis. Results will be reported in aggregate form only — no individual responses will be identifiable in the final thesis.

Participant Information

Participant Name:

Date:

Overall Experiment Experience

1. How would you describe your overall experience during the experiment?

☐ Very positive

☐ Positive

☐ Neutral

☐ Negative

2. Were the instructions and tasks clear throughout both phases?

☐ Yes, entirely

☐ Partially

☐ No

If partially or no, please specify:

Section B - Phase Comparison

3. Which phase did you find less cognitively demanding?

☐ Phase I (without AI)

☐ Phase II (with AI)

☐ Both equally demanding

4. Which lesson plan do you consider more complete or effective?

☐ Phase I lesson plan

☐ Phase II lesson plan

☐ Both equally

5. On what basis are you making this judgment? (Select all that apply)

☐ Quality of content

☐ Alignment with curriculum

☐ Ease of adaptation for class

☐ Personal investment in the plan

☐ Other

6. What differences did you notice between the two lesson plans? Consider aspects such as content coverage, pedagogical approach, adaptation to student level, or your personal investment in the plan. (3–5 sentences recommended, if video is not recorded of this feedback)

Impact of Generative AI on Your Practice

Please rate your agreement with each statement based on your experience in Phase II.

Statement	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Generative AI helped me prepare the lesson plan more effectively.	☐	☐	☐	☐	☐
Using AI reduced the time required for lesson planning.	☐	☐	☐	☐	☐
Using AI reduced the effort required for lesson planning.	☐	☐	☐	☐	☐
My creative or pedagogical ideas changed while working with AI.	☐	☐	☐	☐	☐
The AI-generated content was relevant to my students' level and needs.	☐	☐	☐	☐	☐
I felt in control of the lesson plan even when using AI assistance.	☐	☐	☐	☐	☐

Future Use and Perceived Barriers

7. Would you consider using generative AI tools in your future teaching practice?

☐ Yes, definitely ☐ Probably yes ☐ Unsure ☐ Probably not ☐ No

Please indicate which of the following you perceive as barriers to using AI in your teaching:

Potential Barrier	Yes	No	Unsure
Institutional restrictions on AI use in schools	☐	☐	☐
Lack of training or knowledge about AI tools	☐	☐	☐
Ethical concerns about AI-generated content	☐	☐	☐
Uncertainty about pedagogical added value	☐	☐	☐
Concerns about student data privacy	☐	☐	☐
Lack of reliable internet access or devices	☐	☐	☐

8. If you identified barriers above, which do you consider the most significant, and why? (3–5 sentences recommended, if video is not recorded of this feedback)

Additional Comments

9. Is there anything else you would like to share about your experience, the experiment design, or the use of AI in teaching?

Thank you for your participation. Your feedback is valuable to this research.